

第二章习题

Lemma: 任取 $s \in \mathbb{Z}_{\geq 2}$. 对 $\forall n \in \mathbb{Z}_{\geq 0}$ 都有唯一的 s 进制展开:

$$n = a_0 + a_1 s + a_2 s^2 + \dots, \quad a_k \in \{0, 1, \dots, s-1\}$$

其中至多仅有有限个 a_k 非零.

Proof: 当 $n=0$ 时, $n=0=0+0 \cdot s+0 \cdot s^2+0 \cdot s^3+\dots$

当 $n=1$ 时, $n=1=1+0 \cdot s+0 \cdot s^2+0 \cdot s^3+\dots$

对 $\forall n \in \mathbb{Z}_{\geq 2}$. 若对 $\forall y \in \mathbb{Z}_{\geq 0}$ 且 $y < n$, 都有: y 有唯一的 s 进制展开.

则有: $\because n \in \mathbb{Z}_{\geq 2}, s \in \mathbb{Z}_{\geq 2} \quad \therefore \exists$ 唯一的 $q, r \in \mathbb{Z}$, s.t.

$$n = sq + r, \quad 0 \leq r < |s| = s$$

$$\therefore n = r + sq, \quad r \in \{0, 1, \dots, s-1\}$$

$$\because r \geq 0 \quad \therefore -r \leq 0 \quad \therefore sq = n - r = n + (-r) \leq n \quad \therefore q \leq \frac{n}{s} < n.$$

假设 $q < 0$, 则有: $q \leq -1 \quad \therefore sq \leq -s$

$\therefore n = sq + r \leq -s + r < -s + s = 0$. 这与 $n \in \mathbb{Z}_{\geq 0}$ 矛盾.

$$\therefore q \geq 0 \quad \therefore q \in \mathbb{Z}_{\geq 0}.$$

$\therefore q \in \mathbb{Z}_{\geq 0}$ 且 $q < n \quad \therefore q$ 有唯一的 s 进制展开.

$$\text{设 } q = \alpha_0 + \alpha_1 s + \alpha_2 s^2 + \dots + \alpha_t s^t \quad (\alpha_{t+1} = \alpha_{t+2} = \dots = 0)$$

$$\therefore n = r + sq = r + \alpha_0 s + \alpha_1 s^2 + \alpha_2 s^3 + \dots + \alpha_t s^{t+1}$$

$\therefore n$ 有 s 进制展开.

$\therefore$ 对 $\forall n \in \mathbb{Z}_{\geq 0}$, n 有 s 进制展开. $\therefore$ 存在性得证.

下证唯一性. 假设 $n \in \mathbb{Z}_{\geq 0}$ 有 2 个 s 进制展开:

$$n = a_0 + a_1 s + a_2 s^2 + \dots + a_\lambda s^\lambda, \quad a_k \in \{0, 1, \dots, s-1\}$$

$$n = b_0 + b_1 s + b_2 s^2 + \dots + b_\mu s^\mu, \quad b_k \in \{0, 1, \dots, s-1\}$$

分情况讨论如下:

① $\lambda = \mu$. 则有: $a_0 + a_1 s + a_2 s^2 + \dots + a_\lambda s^\lambda = b_0 + b_1 s + b_2 s^2 + \dots + b_\lambda s^\lambda$

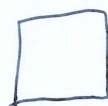
$$\therefore (a_0 - b_0) + (a_1 - b_1)s + (a_2 - b_2)s^2 + \dots + (a_\lambda - b_\lambda)s^\lambda = 0$$

我暂时不知道 ~~怎么~~ 怎么证明 $a_0 - b_0 = a_1 - b_1 = a_2 - b_2 = \dots = a_\lambda - b_\lambda = 0$.

② $\lambda > \mu$ ③ $\lambda < \mu$

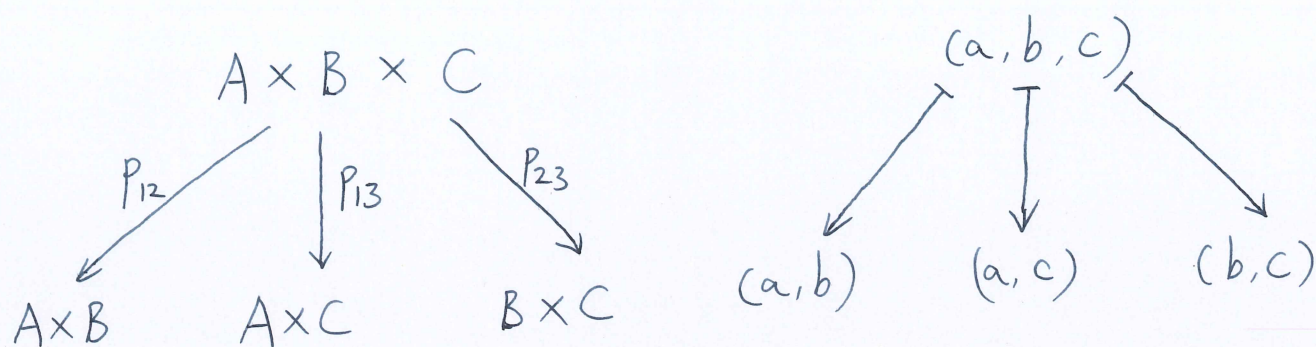
暂时不会证.

目前只证明了存在性, 还不会证唯一性.



第二章 末尾习题

第2题. 设 A, B, C 为集合. 定义所谓的坐标投影映射:



说明当映射 $f: A \rightarrow B$ 和 $g: B \rightarrow C$ 给定, $gf: A \rightarrow C$ 的函数图形 Γ_{gf} 是:

$$\Gamma_{gf} = p_{13} (p_{12}^{-1}(\Gamma_f) \cap p_{23}^{-1}(\Gamma_g))$$

Proof: $\because f: A \rightarrow B$ 是映射 $\therefore \Gamma_f = \{(a, f(a)) : a \in A\} \subseteq A \times B$

$\because g: B \rightarrow C$ 是映射 $\therefore \Gamma_g = \{(b, g(b)) : b \in B\} \subseteq B \times C$

$\because f: A \rightarrow B$ 是映射, $g: B \rightarrow C$ 是映射 $\therefore gf: A \rightarrow C$ 是映射.

$\therefore \Gamma_{gf} = \{(a, c) \in A \times C : \exists b \in B, \text{使得 } (a, b) \in \Gamma_f \text{ 且 } (b, c) \in \Gamma_g\} \subseteq A \times C$

$\because p_{12}: A \times B \times C \rightarrow A \times B$ 是映射, $\Gamma_f \subseteq A \times B$

$\therefore p_{12}^{-1}(\Gamma_f) = \{(a, b, c) \in A \times B \times C : (a, b) \in \Gamma_f\} \subseteq A \times B \times C$

$\because p_{23}: A \times B \times C \rightarrow B \times C$ 是映射, $\Gamma_g \subseteq B \times C$

$\therefore p_{23}^{-1}(\Gamma_g) = \{(a, b, c) \in A \times B \times C : (b, c) \in \Gamma_g\} \subseteq A \times B \times C$

对 $\forall (a, c) \in \text{左边}$, 有: $(a, c) \in \Gamma_{gf}$

$\therefore \exists b \in B$, s.t. $(a, b) \in \Gamma_f$ 且 $(b, c) \in \Gamma_g$

$\therefore (a, b, c) \in p_{12}^{-1}(\Gamma_f)$ 且 $(a, b, c) \in p_{23}^{-1}(\Gamma_g)$

$$\therefore (a, b, c) \in P_{12}^{-1}(\Gamma_f) \cap P_{23}^{-1}(\Gamma_g)$$

$$\therefore (a, c) = P_{13}((a, b, c)) \in P_{13}(P_{12}^{-1}(\Gamma_f) \cap P_{23}^{-1}(\Gamma_g))$$

$$\therefore (a, c) \in \text{右边} \quad \therefore \text{左边} \subseteq \text{右边}$$

$$\text{又} \forall (a, c) \in \text{右边}, \text{有: } (a, c) \in P_{13}(P_{12}^{-1}(\Gamma_f) \cap P_{23}^{-1}(\Gamma_g)) \subseteq A \times C$$

$$\therefore \exists (\alpha, \beta, \gamma) \in P_{12}^{-1}(\Gamma_f) \cap P_{23}^{-1}(\Gamma_g), \text{ s.t. } P_{13}((\alpha, \beta, \gamma)) = (a, c)$$

$$\therefore (\alpha, \gamma) = (a, c) \quad \therefore a = \alpha, c = \gamma$$

$$\therefore (\alpha, \beta, \gamma) \in P_{12}^{-1}(\Gamma_f) \quad \therefore (\alpha, \beta) \in \Gamma_f \quad \therefore (a, \beta) \in \Gamma_f$$

$$\therefore (\alpha, \beta, \gamma) \in P_{23}^{-1}(\Gamma_g) \quad \therefore (\beta, \gamma) \in \Gamma_g \quad \therefore (\beta, c) \in \Gamma_g$$

$$\therefore \exists \beta \in B, \text{ s.t. } (a, \beta) \in \Gamma_f \text{ 且 } (\beta, c) \in \Gamma_g$$

$$\therefore (a, c) \in \Gamma_{gf} \quad \therefore (a, c) \in \text{左边} \quad \therefore \text{右边} \subseteq \text{左边}$$

$$\therefore \text{左边} = \text{右边} \quad \square$$

第3题. 给出自然的双射 $A^{I \times J} \xrightarrow{1:1} (A^I)^J$, 其中 I, J, A 是任意集合.

解: 定义映射 $\varphi: A^{I \times J} \longrightarrow (A^I)^J$ 为:

$$\varphi: A^{I \times J} \longrightarrow (A^I)^J$$

$$f \longmapsto \varphi(f): J \longrightarrow A^I$$

$$j \longmapsto \begin{matrix} I \longrightarrow A \\ i \longmapsto f(i, j) \end{matrix}$$

对 $\forall f \in A^{I \times J}$ 首先证明 $\varphi(f): J \longrightarrow A^I$ 确实是 $J \rightarrow A^I$

$$j \longmapsto \begin{matrix} I \rightarrow A \\ i \mapsto f(i, j) \end{matrix}$$

的映射.

对 $\forall j \in J$, $\varphi(f)(j): I \rightarrow A$ 是 $I \rightarrow A$ 的映射.
 $i \mapsto f(i, j)$

对 $\forall j_1, j_2 \in J$, 若 $j_1 = j_2$, 则有:

$$\varphi(f)(j_1): I \rightarrow A \quad \begin{matrix} i \mapsto f(i, j_1) \end{matrix} = \varphi(f)(j_2): I \rightarrow A \quad \begin{matrix} i \mapsto f(i, j_2) \end{matrix}$$

$\therefore \varphi(f)$ 确实是 $J \rightarrow A^I$ 的映射.

对 $\forall f_1, f_2 \in A^{I \times J}$ 若 $f_1 = f_2$, 则有:

$$\varphi(f_1): J \longrightarrow A^I \quad \begin{matrix} j \mapsto \begin{matrix} I \rightarrow A \\ i \mapsto f_1(i, j) \end{matrix} \end{matrix} = \varphi(f_2): J \longrightarrow A^I \quad \begin{matrix} j \mapsto \begin{matrix} I \rightarrow A \\ i \mapsto f_2(i, j) \end{matrix} \end{matrix}$$

~~在群~~ $\varphi(f_1) = \varphi(f_2) \Leftrightarrow$ 对 $\forall j \in J$, $\varphi(f_1)(j) = \varphi(f_2)(j)$

$\therefore \varphi$ 是 $A^{I \times J} \longrightarrow (A^I)^J$ 的映射.

对 $\forall f_1, f_2 \in A^{I \times J}$ 若 $\varphi(f_1) = \varphi(f_2)$, 则有:

对 $\forall (i, j) \in I \times J$.

$$\therefore \varphi(f_1) = \varphi(f_2), j \in J \quad \therefore \varphi(f_1)(j) = \varphi(f_2)(j)$$

$$\begin{aligned} \therefore \varphi(f_1)(j) : I &\longrightarrow A &= & \varphi(f_2)(j) : I \longrightarrow A \\ v &\longmapsto f_1(v, j) & & v \longmapsto f_2(v, j) \end{aligned}$$

$$\therefore v \in I \quad \therefore \varphi(f_1)(j)(v) = \varphi(f_2)(j)(v)$$

$$\therefore f_1(v, j) = f_2(v, j) \quad \therefore f_1 = f_2$$

$\therefore \varphi$ 是 $A^{I \times J} \longrightarrow (A^I)^J$ 的单射.

对 $\forall h \in (A^I)^J$, 有: h 是 $J \rightarrow A^I$ 的映射.

$$\begin{aligned} \text{定义映射 } f : I \times J &\longrightarrow A \\ (v, j) &\longmapsto h(j)(v) \end{aligned}$$

则有: $\varphi(f)$ 是 $J \rightarrow A^I$ 的映射.

$$\begin{aligned} \text{对 } \forall j \in J, \quad \varphi(f)(j) : I &\longrightarrow A & h(j) : I &\longrightarrow A \\ v &\longmapsto f(v, j) & v &\longmapsto h(j)(v) \end{aligned}$$

$$\therefore \varphi(f)(j) = h(j) \quad \therefore \varphi(f) = h$$

$\therefore \varphi$ 是 $A^{I \times J} \longrightarrow (A^I)^J$ 的满射

$\therefore \varphi$ 是 $A^{I \times J} \longrightarrow (A^I)^J$ 的双射. □